



# The ProbLemma's Channel Season 3 Guide

## Season 3 Episode 1: The Index Of Tools

- All the problem-solving approaches studied so far, across seasons 1 and 2, are indexed by the episodes in which they were defined and practiced

## Season 3 Episode 2: An Integral Transformation

- Problem **S3M1** solved, show that for all  $\theta$ ,  $x$  real, with  $x > 1$ , and  $n = 1, 2, 3, 4, 5 \dots$  it is the case that:

$$\int_0^\pi \left( x + \sqrt{x^2 - 1} \cos(\theta) \right)^n d\theta = \int_0^\pi \frac{d\theta}{\left( x - \sqrt{x^2 - 1} \cos(\theta) \right)^{n+1}}$$

- Problems **S3M2/3/4** formulated:
  - Problems **S3M2/3/4**: deduce in at least 3 distinct ways an alternative expression (a product) for the following finite sum:

$$\sum_{k=0}^n \cos(a + k\theta)$$

## Season 3 Episode 3: One Telescope, Two Telescopes

- Problem **S3M2** solved

## Season 3 Episode 4: A Complex Approach

- Problem **S3M3** solved

## Season 3 Episode 5: Euclid Says

- Problems **S3M4** solved
- Problem **S3P1** formulated:
  - **S3P1**: find the period of oscillation of a triangular spring oscillator

## Season 3 Episode 6: Triangular Spring Oscillator

- Problem **S3P1** solved
- Problem **S3CS1** formulated:
  - **S3CS1**: sort 3 integers creatively, without the popular containers and/or industrial strength algorithms

## Season 3 Episode 7: 3-Integer Sort, A Discovery

- Problem **S3CS1** solved

- Problem S3M5 formulated:
  - S3M5: construct a point, N, inside of a right triangle, ABC, such that the magnitudes of all the angles NAB, NBC and NCA are all equal one another

#### Season 3 Episode 8: Brocard Points Introduction

- Problem S3M5 solved
- Problem S3M6 formulated:
  - S3M6: invent a way to evaluate finite sums based on a popular approach that is used in combinatorial analysis to prove binomial identities

#### Season 3 Episode 9: Summation By Double-Counting

- Problem S3M6 solved
- Problem S3P2 formulated:
  - S3P2: find the distance covered by a bouncing material point

#### Season 3 Episode 10: Infinity In Physics

- Problem S3P2 solved
- Problem S3CS2 formulated:
  - S3CS2: suggest a Turing Machine-friendly algorithm for testing if a given point is located outside or inside of a given triangle

#### Season 3 Episode 11: A Point-in-triangle Detection Algorithm

- Problem S3CS2 solved
- Problem S3M7 formulated:
  - S3M7: find a way to integrate a binomial differential  $x^m(a + bx^n)^p dx$

#### Season 3 Episode 12: Integration Of Binomial Differentials

- Problem S3M7 solved
- Problem S3M8 formulated:
  - S3M8: use the integration of binomial differentials machinery developed in the previous problem in order to evaluate the following infinite sum:

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+1)(n+2)\dots(n+p)}, \quad p \geq 2$$

#### Season 3 Episode 13: Binomial Differentials Slay Infinite Sums Before Breakfast

- Problem S3M8 solved
- Problem S3P3 formulated:
  - S3P3: distinguish a sphere from an identical ball of same radius and mass

### Season 3 Episode 14: Sisyphus Versus Spherical Chickens Of Uniform Density

- Problem S3P3 solved
- Problem S3CS3 formulated:
  - S3CS3: construct a single-pass algorithm for sorting 0s and 1s

### Season 3 Episode 15: Single-pass Janus Sort

- Problem S3CS3 solved
- Problem S3M9 formulated:
  - S3M9: construct an equilateral triangle on 3 parallel straight lines

### Season 3 Episode 16: An Equilateral Triangle On 3 Parallel Straight Lines Construction

- Problem S3M9 solved
- Problem S3M10 formulated:
  - S3M10: reconstruct a square given 4 points that belong to one side of that square each

### Season 3 Episode 17: 4-Point Square Reconstruction

- Problem S3M10 solved
- Problem S3P4 formulated:
  - S3P4: find the normal and the tangential accelerations as functions of time of a point that moves along a parabola with its acceleration vector staying parallel to the  $y$ -axis at all times

### Season 3 Episode 18: A Parabolic Motion With Acceleration Staying Parallel To The $y$ -axis

- Problem S3P4 solved
- Problem S3CS4 formulated:
  - S3CS4: creatively swap 2 sub-arrays

### Season 3 Episode 19: A Creative Swap Of Two Sub-Arrays

- Problem S3CS4 solved
- Problem S3M11 formulated:
  - S3M11: evaluate a finite trigonometric sum

$$\sum_{k=1}^n \frac{1}{\sin(2^k x)}$$

### Season 3 Episode 20: A Finite Trigonometric Sum Evaluation

- Problem S3M11 solved
- Problem S3M12 formulated:
  - S3M12: evaluate a finite binomial sum using Physics:

$$\sum_{k=0}^n k \cdot \binom{n}{k}$$

### Season 3 Episode 21: Physics Evaluates A Finite Binomial Sum

- Problem S3M12 solved
- Problem S3P5 formulated:
  - S3P5: recover the shape of the trajectory of a point that moves in a plane in such a way that its both tangential and normal accelerations remain constant at all times

### Season 3 Episode 22: Planar Motion With Constant Tangential And Normal Accelerations

- Problem S3P5 solved
- Problem S3CS5 formulated:
  - S3CS5: explain how the Shunting Yard Algorithm works and create a Slow-Motion Player that lets a user experiment with and witness the mechanics of that algorithm

### Season 3 Episode 23: The Mechanics Of The Shunting Yard Algorithm

- Problem S3CS5 solved
- Problem S3M13 formulated:
  - S3M13: evaluate the following integrals in at least 3 different creative ways:

$$C_{a,b}(x) = \int e^{ax} \cos(bx) dx; \quad S_{a,b}(x) = \int e^{ax} \sin(bx) dx$$

### Season 3 Episode 24: 3 Creative Evaluations Of Ubiquitous Integrals

- Problem S3M13 solved
- Problem S3M14 formulated: a divisibility by 7 test
  - S3M14: design a divisibility-by-7 algorithm using the Reduce-And-Conquer approach

### Season 3 Episode 25: A Divisibility-By-7 Test Design Via Reduce-And-Conquer

- Problem S3M14 solved
- Problem S3P6 formulated: a kinematic descend of a rotating line segment
  - S3M14: recover the law of motion, the trajectory, the velocity, the acceleration and the curvature of the trajectory traced by one edge of a uniformly descending line segment that also rotates in the horizontal plane with a constant angular velocity

### Season 3 Episode 26: A Kinematic Descend Of A Rotating Line Segment

- Problem S3P6 solved
- Problem S3CS6 formulated: describe the mechanics and then design and implement the Narayana's Next-Linear-Permutation algorithm

### Season 3 Episode 27: Narayana's Next Linear Permutation Algorithm

- Problem S3CS6 solved
- Problem S3M15 formulated. Evaluate the following finite trigonometric sum:

$$\sum_{m=1}^n \arctan \left( \frac{2m}{m^4 + m^2 + 2} \right)$$

### Season 3 Episode 28: In Like A Lion Out Like A Lamb

- Problem S3M15 solved
- Problem S3M16 formulated. Evaluate the following limit:

$$\lim_{n \rightarrow +\infty} n^{\frac{1}{n}}$$

### Season 3 Episode 29: The Art Of Manipulation Of Inequalities

- Problem S3M16 solved
- Problem S3P7 formulated: deduce an equation that describes a spherical motion of a point under a constant angle with the meridians

### Season 3 Episode 30: A Deduction Of An Equation Of A Loxodrome

- Problem S3P7 solved
- Problem S3CS7 formulated: construct an  $O(n)$  time and  $O(1)$  space complexity algorithm for detecting if a given integer can be represented as a sum of two integers from a given array sorted in some order, ascending or descending

### Season 3 Episode 31: An Integer As A Sum Of Two Elements Of A Sorted Array

- Problem S3CS7 solved
- Problem S3M17 formulated: evaluate the following indefinite integral

$$\int \sqrt{x^2 - a^2} dx$$

in at least two creative ways.

### Season 3 Episode 32: Hyperbolically Speaking

- Problem S3M17 solved
- Problem S3M18 formulated: evaluate the following definite integrals

$$\int_0^{\frac{\pi}{2}} \sin^m(x) dx \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^m(x) dx$$

for  $m = 1, 2, 3, 4, 5$  and so on.

### Season 3 Episode 33: Recurrence Relations Evaluate Integrals

- Problem S3M18 solved
- Problem S3M19 formulated: show that hyperbolic functions can be defined by a direct analogy with circular functions defined via origin-centered unit circle

Season 3 Episode 34: Hyperbolic Functions Defined By A Direct Analogy With Circular Functions

- Problem **S3M19** solved
- Problem **S3P8** formulated: recover the law of motion of a point that is moving along a straight line that sweeps the surface of a right circular cone

Season 3 Episode 35: Twice Telescopic Rotating Antenna, The Kinematics Of

- Problem **S3P8** solved
- Problem **S3CS8** formulated: construct an algorithm for detecting a cycle in a linked list

Season 3 Episode 36: A Vicious Circle Or Lords Of The Ring

- Problem **S3CS8** solved
- Problem **S3M20** formulated: show that for all real  $x \neq \pm k\pi$ , where  $k = 0, 1, 2, 3$  and so on, it is the case that:

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Season 3 Episode 37: Sine As A Product, With Applications

- Problem **S3M20** solved
- Problem **S3M21** formulated: evaluate the Euler-Poisson integral

$$\int_0^{+\infty} e^{-x^2} dx$$

using the Squeeze Theorem.

Season 3 Episode 38: What Mathematicians Do All Day Or Euler-Poisson Integral Evaluation via the Squeeze Theorem

- Problem **S3M21** solved
- Problem **S3P9** formulated: solve the **S2E54/S2P2** problem under the assumption that there is no external input into the rotating rod-bead system, which means that the system's angular velocity is no longer constant.

Season 3 Episode 39: A Bead Sliding Off Of A Massless Rod That Rotates With A Decreasing Angular Velocity

- Problem **S3P9** solved
- Problem **S3CS9** formulated: find the diamond across 8 boxes with 2 cooperating people if boxes have a random orientation, the two people must agree on the algorithm ahead of time, one person is lead into the room with the boxes to do something and then escorted out of the room, while the second person, upon entering that room, has to recover the box with the diamond

Season 3 Episode 40: Computer Science And Group Theory Find A Diamond

- Problem **S3CS9** solved
- Problem **S3M22** formulated: show

Season 3 Episode 41: Lorem Ipsum

- Problem **S3M22** solved
- Problem **S3M23** formulated: show

Season 3 Episode 42: Lorem Ipsum

- Problem **S3M23** solved
- Problem **S3P10** formulated: show

Season 3 Episode 43: Lorem Ipsum

- Problem **S3P10** solved
- Problem **S3CS10** formulated: show

Season 3 Episode 44: Lorem Ipsum

- Problem **S3CS10** solved
- Problem **S3M24** formulated: show

Season 3 Episode 45: Lorem Ipsum

- Problem **S3M24** solved
- Problem **S3M25** formulated: show

Season 3 Episode 46: Lorem Ipsum

- Problem **S3M25** solved
- Problem **S3P11** formulated: show

Season 3 Episode 47: Lorem Ipsum

- Problem **S3P11** solved
- Problem **S3CS11** formulated: show

Season 3 Episode 48: Lorem Ipsum

- Problem **S3CS11** solved
- Problem **S3M26** formulated: show

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